Εθνικό Μετσόβιο Πολυτεχνείο

Σχολή Ηλεκτρολόγων Μηχανικών και Μηχανικών Υπολογιστών

Αναφορά 1ου εργαστηρίου

“Ψηφιακά Συστήματα VLSI”

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Αθήνα

2026

Α' ΜΕΡΟΣ

Άσκηση Α2:

- ***VHDL code***

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity dec8 is
Port (enc: in STD_LOGIC_VECTOR (2 downto 0);
     dec: out STD_LOGIC_VECTOR (7 downto 0)
     );
end dec8;

architecture Behavioral of dec8 is
begin
   process (enc)
   begin
      case (enc) is
         when "111" => dec <= "10000000";
         when "110" => dec <= "01000000";
         when "101" => dec <= "00100000";
         when "100" => dec <= "00010000";
         when "011" => dec <= "00001000";
         when "010" => dec <= "00000100";
         when "001" => dec <= "00000010";
         when "000" => dec <= "00000001";
         when others => dec <= "00000000";
      end case;
   end process;
end Behavioral;

architecture Dataflow of dec8 is
```

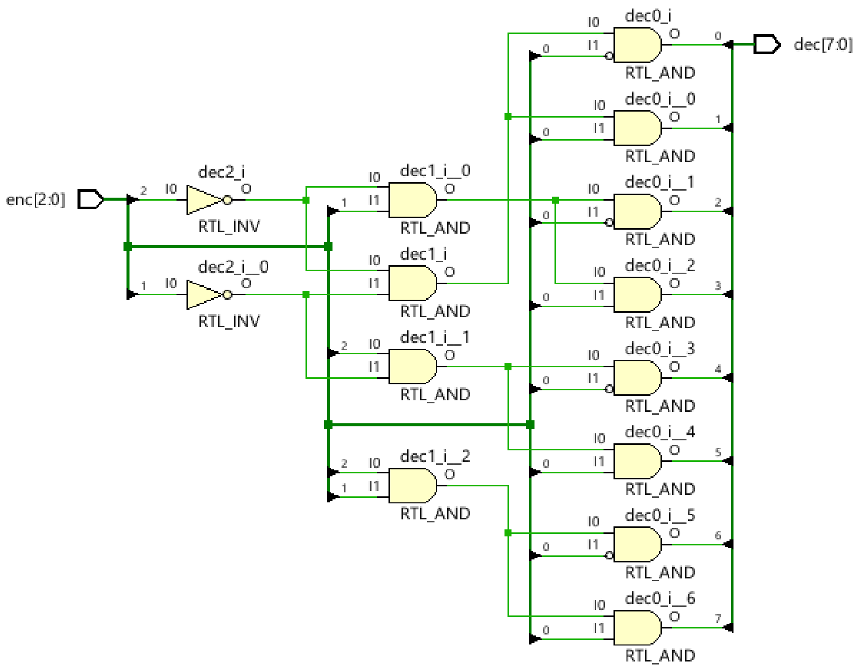
```
begin
   dec(0) <= (not enc(2)) and (not enc(1)) and (not enc(0)); -- 000
   dec(1) <= (not enc(2)) and (not enc(1)) and enc(0);       -- 001
   dec(2) <= (not enc(2)) and enc(1) and (not enc(0));       -- 010
   dec(3) <= (not enc(2)) and enc(1) and enc(0);             -- 011
   dec(4) <= enc(2) and (not enc(1)) and (not enc(0));       -- 100
   dec(5) <= enc(2) and (not enc(1)) and enc(0);             -- 101
   dec(6) <= enc(2) and enc(1) and (not enc(0));             -- 110
   dec(7) <= enc(2) and enc(1) and enc(0);                   -- 111
end Dataflow;
```

- ***Testbench***

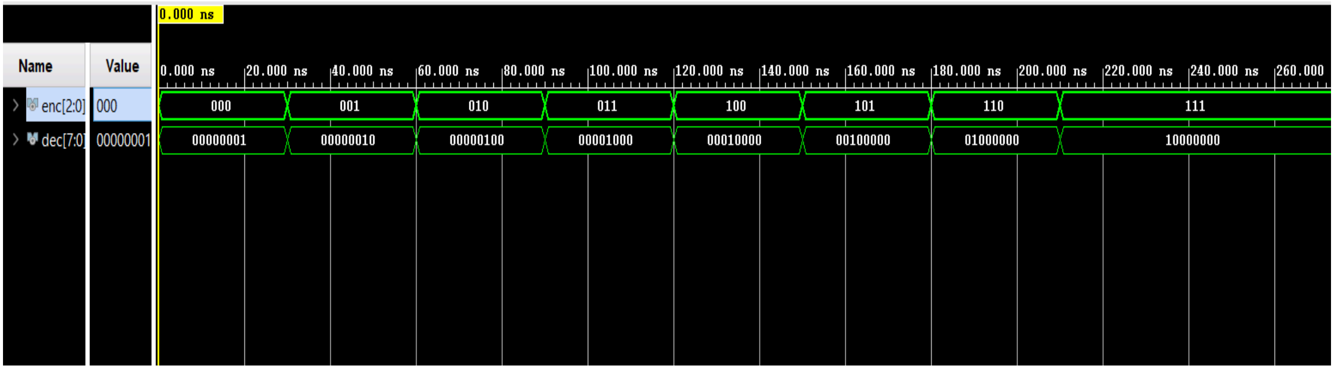
```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity dec_8_tb is
--  Port ( );
end dec_8_tb;

architecture testbench of dec_8_tb is

   signal enc : STD_LOGIC_VECTOR (2 downto 0);
   signal dec : STD_LOGIC_VECTOR (7 downto 0);

begin
   dut: entity work.dec8(Behavioral) port map (  -- in this line we change the Behavioral in the parenthesis to Dataflow to work in the other architecture
      enc => enc,
      dec => dec
   );

   stimulus: process
   begin
```

```
        enc <= "000";
        wait for 30 ns;
        enc <= "001";
        wait for 30 ns;
        enc <= "010";
        wait for 30 ns;
        enc <= "011";
        wait for 30 ns;
        enc <= "100";
        wait for 30 ns;
        enc <= "101";
        wait for 30 ns;
        enc <= "110";
        wait for 30 ns;
        enc <= "111";
        wait for 30 ns;

        wait;

    end process;
end testbench;
```

❖ ***RTL schematic***



- ❖ ***Behavioral Simulation***



<u>Άσκηση B2:</u>

- ❖ ***VHDL code***

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity rshifter_reg3 is
   port (
      clk,rst,si,en,pl,direction: in std_logic;
      din: in std_logic_vector(3 downto 0);
      so: out std_logic);
end rshifter_reg3;


architecture Behavioral of rshifter_reg3 is
   signal dff: std_logic_vector(3 downto 0);
begin
   process (clk,rst)
      begin
         if rst='0' then
```

```
            dff<=(others=>'0');
           elsif clk'event and clk='1' then
           if pl='1' then
            dff<=din;
           elsif en='1' then
            case (direction) is
               when '0' => dff <= si&dff (3 downto 1);
               when '1' => dff <= dff(2 downto 0)&si;
               when others => dff <= "0000";
               end case;
           end if;
       end if;
   end process;
   with direction select
       so <= dff(0) when '0',
             dff(3) when '1',
             '0' when others;
end Behavioral;
```

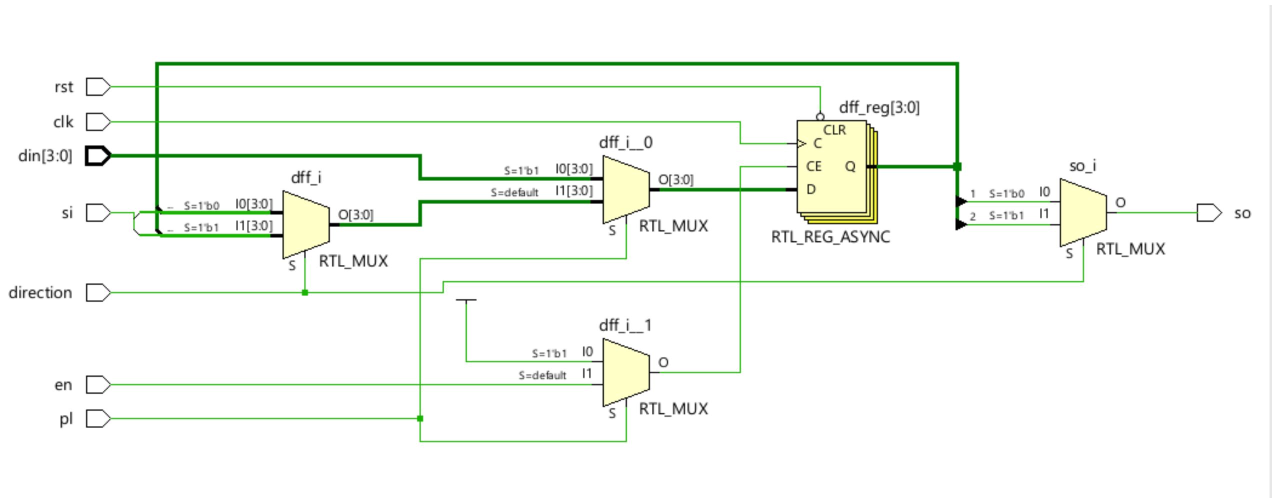
❖ ***Testbench***

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity rshift_reg3_tb is
--  Port ( );
end rshift_reg3_tb;

architecture testbench of rshift_reg3_tb is

    constant clock_Delay : time := 10ns;

    signal clk : std_logic := '0';
    signal rst : std_logic := '0';
```

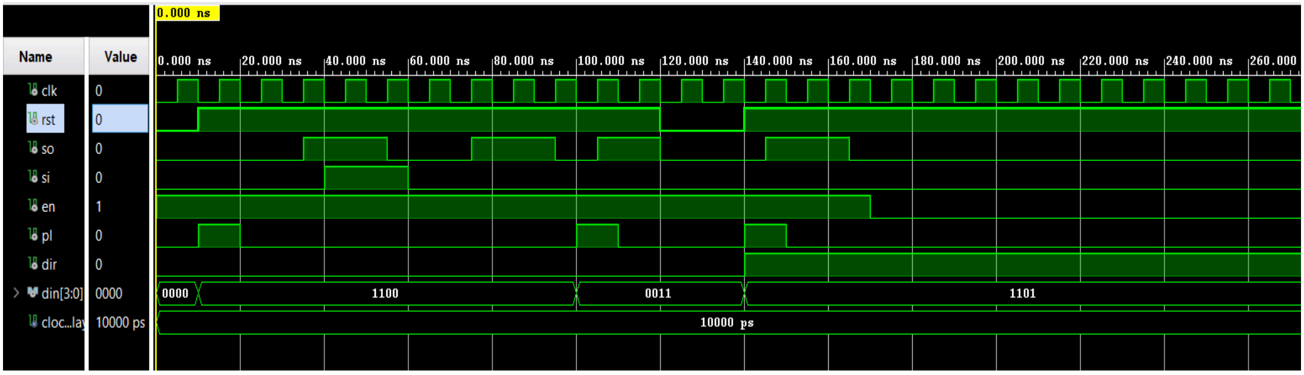
```
    signal so  : std_logic := '0';
    signal si  : std_logic := '0';
    signal en  : std_logic := '0';
    signal pl  : std_logic := '0';
    signal dir : std_logic := '0';
    signal din : std_logic_vector(3 downto 0) := (others => '0');


begin
    dut: entity work.rshifter_reg3(Behavioral)
        port map(clk => clk, rst => rst,
        en=>en,pl=>pl,direction=>dir,din=>din,so=>so,si=>si);

    stimulus: process
    begin
    -- right dir
        en <= '1';

        din <= "0000";
        si <= '0';
        dir <= '0';
        rst <= '0';
        wait for (clock_delay);
        rst <= '1';

        din <= "1100";
        pl <= '1';
        wait for (clock_delay);
        pl <= '0';

        wait for (2*clock_delay);
        si <= '1';
        wait for (2*clock_delay);
        si <= '0';
        wait for (4*clock_delay);
        din <= "0011";
        pl <= '1';
        wait for (clock_delay);
        pl <= '0';
        wait for (clock_delay);
        rst <= '0';
        wait for (2*clock_delay);
        rst <= '1';
    --left dir
```

```
        dir <= '1';
        din <= "1101";
        pl <= '1';
        wait for (clock_delay);
        pl <= '0';
        wait for (2*clock_delay);
        en <= '0';
        wait for (4*clock_delay);

        wait;
    end process stimulus;

    gen_clock: process
    begin
        clk <='0';
        wait for (clock_delay/2);
        clk <='1';
        wait for (clock_delay/2);
    end process gen_clock;

end testbench;
```

❖ ***RTL schematic***



❖ ***Behavioral simulation***



Άσκηση B3(no limit):

❖ ***VHDL code***

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.std_logic_unsigned.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity count3 is
    port(clk,
         resetn,
         count_en : in std_logic ;
         sum : out std_logic_vector(2 downto 0);
         cout : out std_logic ;
         dir : in std_logic);
end count3;

architecture Behavioral of count3 is
        signal count : std_logic_vector(2 downto 0);
begin
```

```
    process(clk, resetn)
    begin
        if resetn='0' then
         count <= (others=>'0');
        elsif rising_edge(clk) then
         if count_en= '1' then
          case dir is
            when '1' =>
                if count /= 7 then
                    count <= count + 1;
                else
                    count <= "000";
                end if;
            when '0' =>
                if count /= 0 then
                    count <= count - 1;
                else
                    count <= "111";
                end if;
            when others =>
                count <= count;
          end case;
        end if;
        end if;
    end process;
    sum<= count;
    cout<= '1' when count=7 and count_en='1' else '0';
end Behavioral;
```

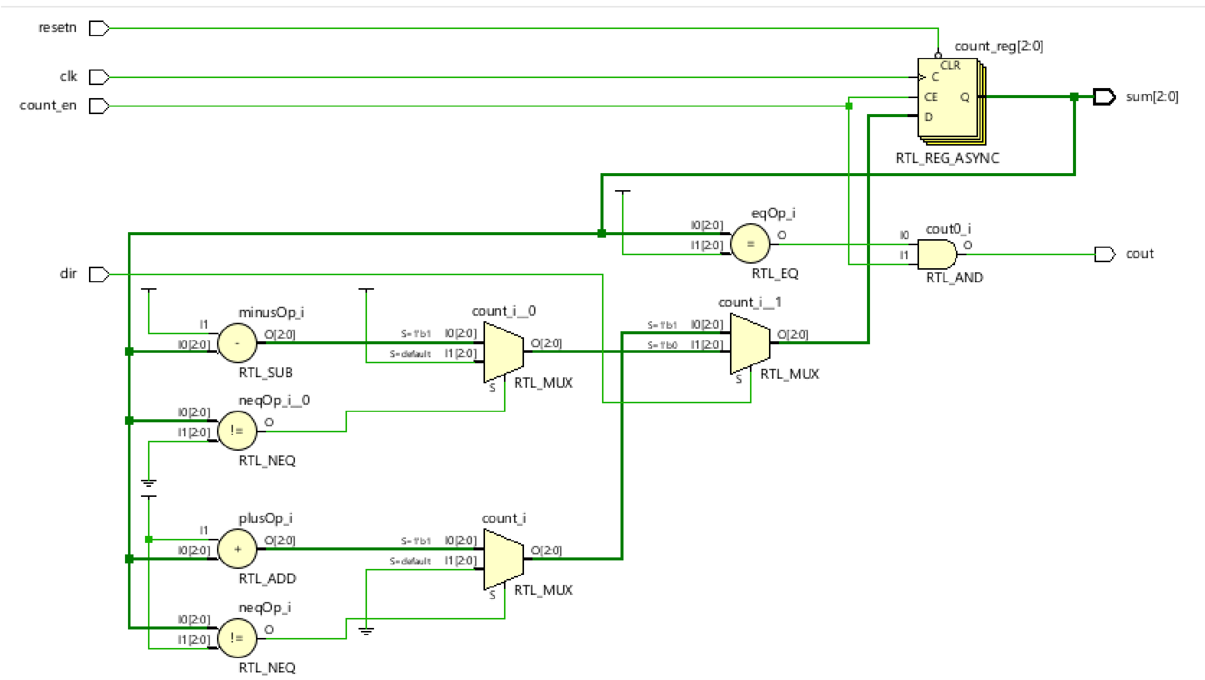
- ***Testbench***

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity count3_tb is
```

```
--  Port ( );
end count3_tb;

architecture testbench of count3_tb is
   signal sum: std_logic_vector(2 downto 0):= (others => '0');
   signal clk,resetn,count_en,dir: std_logic := '0';
   signal cout: std_logic;
   constant clock_delay: time:=10ns;

   begin
      dut: entity work.count3 (behavioral)
      port map(clk=>clk,
             resetn=>resetn,
             count_en=>count_en,
             dir=>dir,
             sum=>sum,
             cout=>cout);
      stimulus: process
      begin
         -- reset
         resetn <= '0';
         wait for clock_delay;
         resetn <= '1';

         count_en <= '1';

         -- count up
         dir <= '1';
         wait for 32*clock_delay;
         wait for 16*clock_delay;

         -- count down
         dir <= '0';
         wait for 32*clock_delay;
         wait for 16*clock_delay;

         -- stop counter
         count_en <= '0';
         wait for 8*clock_delay;

         wait;

         end process;
```

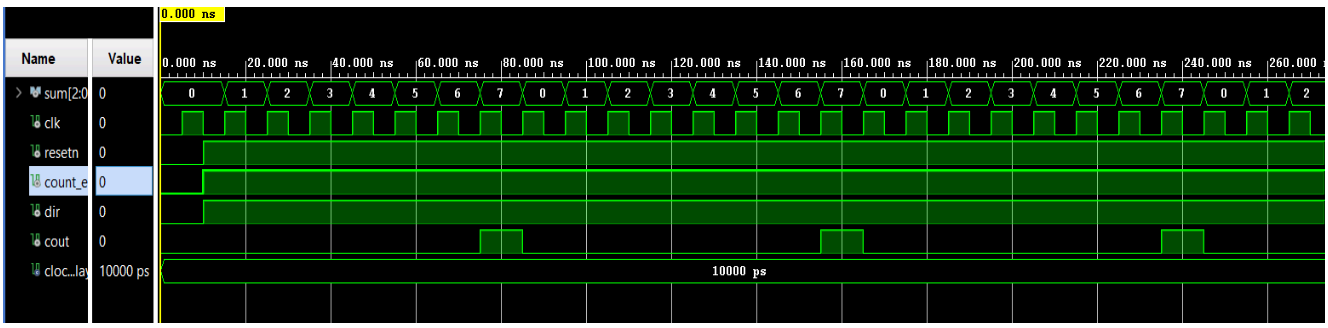
```
        gen_clk: process
        begin
           clk <= '0';
           wait for clock_delay/2;
           clk <= '1';
           wait for clock_delay/2;
        end process;

end testbench;
```

- ***RTL schematic***

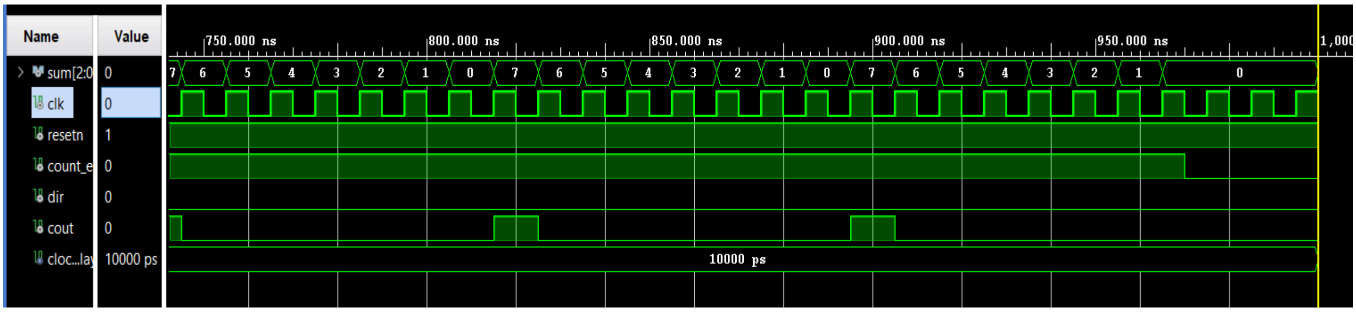


- ***Behavioral simulation***

count up:



count down:



Άσκηση Β3(no limit):

- ***VHDL code***

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.std_logic_unsigned.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity count3_limit is
port(
   clk,
   resetn,
   count_en : in std_logic;
   limit : in std_logic_vector(2 downto 0);
   sum : out std_logic_vector(2 downto 0);
   cout : out std_logic
);
end count3_limit;

architecture Behavioral of count3_limit is
   signal count : std_logic_vector(2 downto 0);
begin
   process(clk, resetn)
```

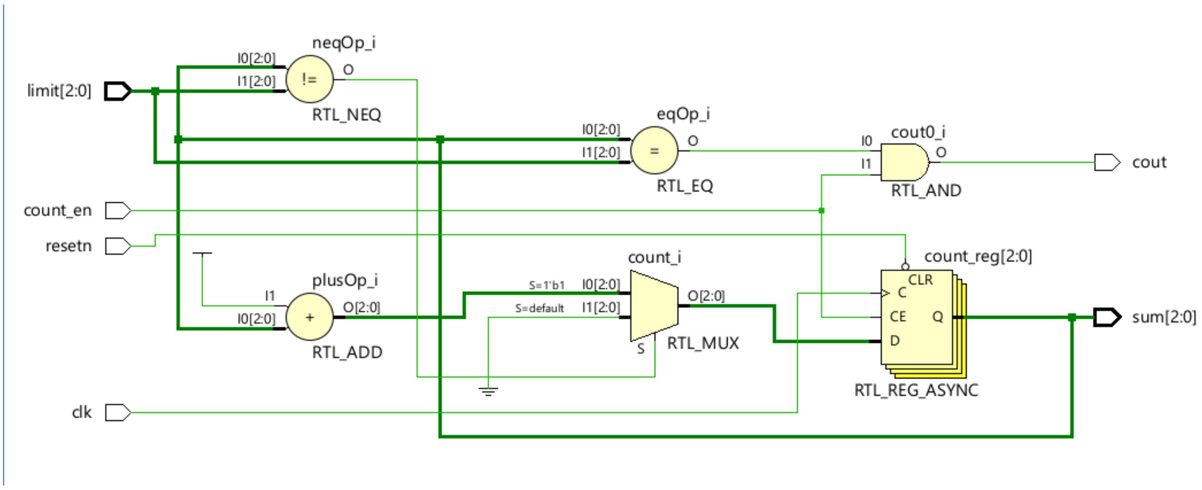
```
    begin

        if resetn='0' then
            count <= (others=>'0');
        elsif clk'event and clk='1' then
            if count_en='1' then
                if count /= limit then
                    count <= count + 1;
                else
                    count <= (others=>'0');
                end if;
            end if;
        end if;
        end process;
        sum <= count;
        cout <= '1' when count = limit and count_en='1' else '0';
end Behavioral;
```

- ***Testbench***

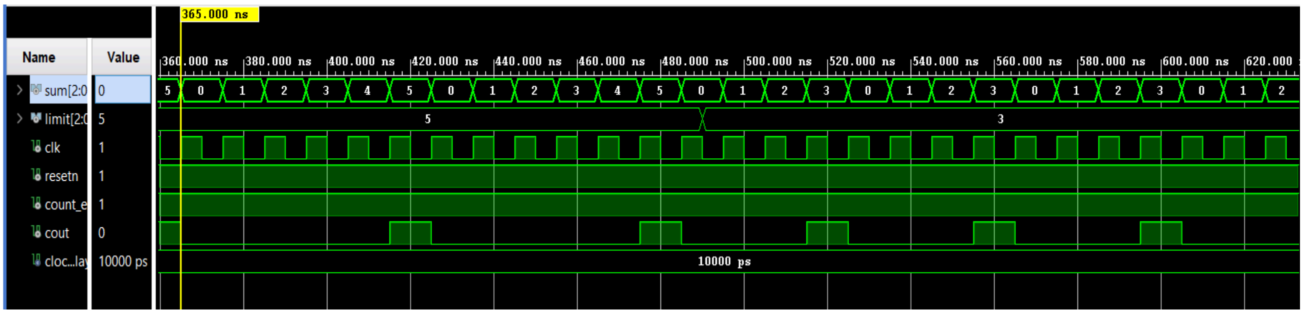
```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity count3_limit_tb is
--  Port ( );
end count3_limit_tb;

architecture testbench of count3_limit_tb is

signal sum : std_logic_vector(2 downto 0);
signal limit : std_logic_vector(2 downto 0) := "101"; -- αρχικό modulo
signal clk : std_logic := '0';
signal resetn : std_logic := '0';
signal count_en : std_logic := '0';
signal cout : std_logic;
```

```
constant clock_delay : time := 10 ns;

begin
    dut: entity work.count3_limit
    port map(
        clk => clk,
        resetn => resetn,
        count_en => count_en,
        limit => limit,
        sum => sum,
        cout => cout
    );

    stimulus: process
    begin
        resetn <= '0';
        wait for clock_delay;
        resetn <= '1';
        count_en <= '1';
        -- TEST modulo = 5
        limit <= "101";
        wait for 32*clock_delay;
        wait for 16*clock_delay;
        -- TEST modulo = 3
        limit <= "011";
        wait for 32*clock_delay;
        wait for 16*clock_delay;
        count_en <= '0';
        wait for 5*clock_delay;
        wait;
    end process;

    gen_clk: process
        begin
            clk <= '0';
            wait for clock_delay/2;
            clk <= '1';
            wait for clock_delay/2;
    end process;
end testbench;
```

- ❖ ***RTL schematic***



- ❖ ***Behavioral simulation***



<u>***B' ΜΕΡΟΣ***</u>

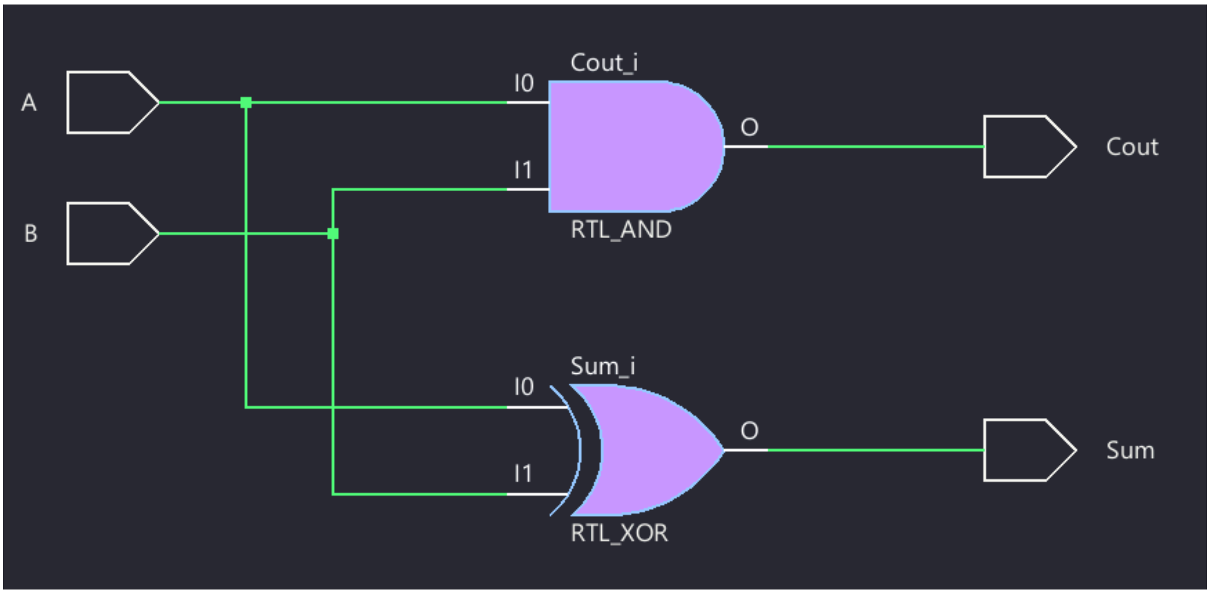
<u>***Half Adder:***</u>

- ❖ ***VHDL Code***

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity half_adder is
   port ( A, B: in std_logic;
          Cout, Sum: out std_logic);
end half_adder;

architecture Dataflow of half_adder is
begin
   Sum <= A xor B;
```

```
    Cout <= A and B;
end Dataflow;
```

- ***RTL Schematic***

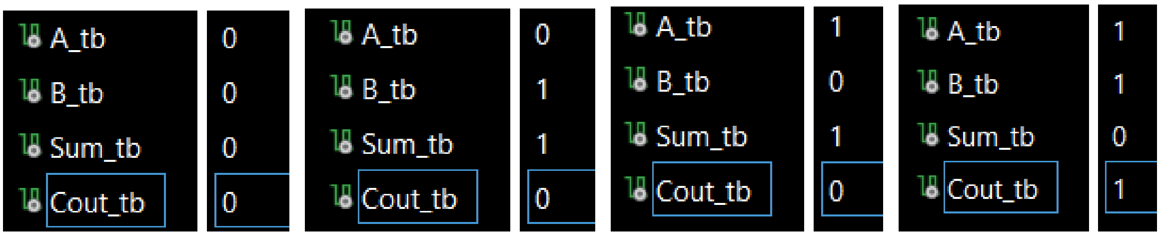


- ***Testbench***

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity half_adder_tb is
end half_adder_tb;

architecture Behavioral of half_adder_tb is
    component half_adder is
        port ( A, B: in std_logic;
               Cout, Sum: out std_logic);
    end component half_adder;

    signal A_tb, B_tb, Sum_tb, Cout_tb: std_logic;
begin
    DUT: half_adder
        port map (A => A_tb,
                  B => B_tb,
                  Sum => Sum_tb,
                  Cout => Cout_tb);
    process
    begin
```

```
        A_tb <= '0'; B_tb <= '0'; wait for 20ns;
        A_tb <= '0'; B_tb <= '1'; wait for 20ns;
        A_tb <= '1'; B_tb <= '0'; wait for 20ns;
        A_tb <= '1'; B_tb <= '1'; wait for 20ns;
        wait; -- only run once
    end process;
end Behavioral;
```

Τα αποτελέσματα της προσομοίωσης επαληθεύουν την ορθή λειτουργία του προγράμματος.



Με την εντολή:
report_timing -from [all_inputs] -to [all_outputs] -max_paths 10
λαμβάνουμε το κρίσιμο μονοπάτι και την χρονική καθυστέρηση που προκαλεί:

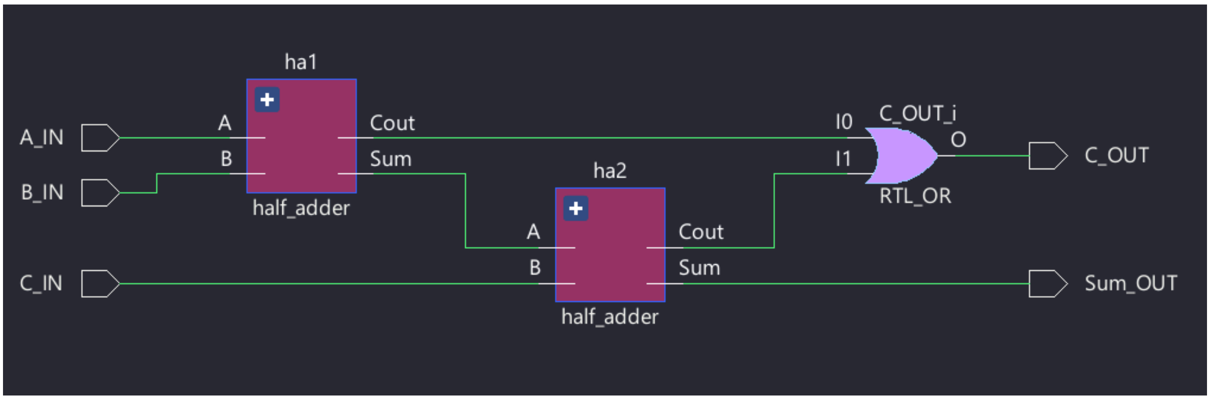
```
Timing Report

Slack:                    inf
  Source:                 B
                            (input port)
  Destination:            Cout
                            (output port)
  Path Group:             (none)
  Path Type:              Max at Slow Process Corner
  Data Path Delay:        7.058ns  (logic 3.972ns (56.269%)  route 3.087ns (43.731%))
  Logic Levels:           3  (IBUF=1 LUT2=1 OBUF=1)

    Location             Delay type                Incr(ns)  Path(ns)    Netlist Resource(s)
  -------------------------------------------------------------------    -------------------
    P10                                               0.000     0.000 r  B (IN)
                         net (fo=0)                   0.000     0.000    B
    P10                  IBUF (Prop_ibuf_I_O)         0.966     0.966 r  B_IBUF_inst/O
                         net (fo=2, routed)           1.366     2.332    B_IBUF
    SLICE_X43Y1          LUT2 (Prop_lut2_I1_O)        0.152     2.484 r  Cout_OBUF_inst_i_1/O
                         net (fo=1, routed)           1.721     4.205    Cout_OBUF
    P9                   OBUF (Prop_obuf_I_O)         2.853     7.058 r  Cout_OBUF_inst/O
                         net (fo=0)                   0.000     7.058    Cout
    P9                                                                r  Cout (OUT)
  -------------------------------------------------------------------    -------------------
```

Full Adder:

- ***VHDL Code***

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity full_adder is
   port ( A_IN, B_IN, C_IN: in std_logic;
          C_OUT, Sum_OUT: out std_logic);
end full_adder;

architecture Structural of full_adder is
   component half_adder is
      port ( A, B: in std_logic;
             Cout, Sum: out std_logic);
   end component;

   signal temp_sum, temp_carry, int_signal: std_logic;

begin
   ha1: half_adder port map (A => A_IN,
                             B => B_IN,
                             Cout => temp_carry,
                             Sum => temp_sum);

   ha2: half_adder port map (A => temp_sum,
                             B => C_IN,
                             Cout => int_signal,
                             Sum => Sum_OUT);

   C_OUT <= temp_carry or int_signal;

end Structural;
```

❖ *RTL Schematic*



❖ *Testbench*

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity full_adder_tb is
end full_adder_tb;

architecture Behavioral of full_adder_tb is
   component full_adder is
      port ( A, B, Cin: in std_logic;
             Cout, Sum: out std_logic);
   end component full_adder;

   signal A_tb, B_tb, Cin_tb, Sum_tb, Cout_tb: std_logic;
begin
   DUT: full_adder    -- DUT: device under test
      port map (A => A_tb,
               B => B_tb,
               Cin => Cin_tb,
               Sum => Sum_tb,
               Cout => Cout_tb);
   process
   begin
      A_tb <= '0'; wait for 160ns;
      A_tb <= '1'; wait for 160ns;
   end process;

   process
   begin
      B_tb <= '0'; wait for 80ns;
      B_tb <= '1'; wait for 80ns;
   end process;
```

```
    process
    begin
        Cin_tb <= '0'; wait for 40ns;
        Cin_tb <= '1'; wait for 40ns;
    end process;
end Behavioral;
```

Τα αποτελέσματα της προσομοίωσης επαληθεύουν την ορθή λειτουργία του προγράμματος.

- ***Timing Report***

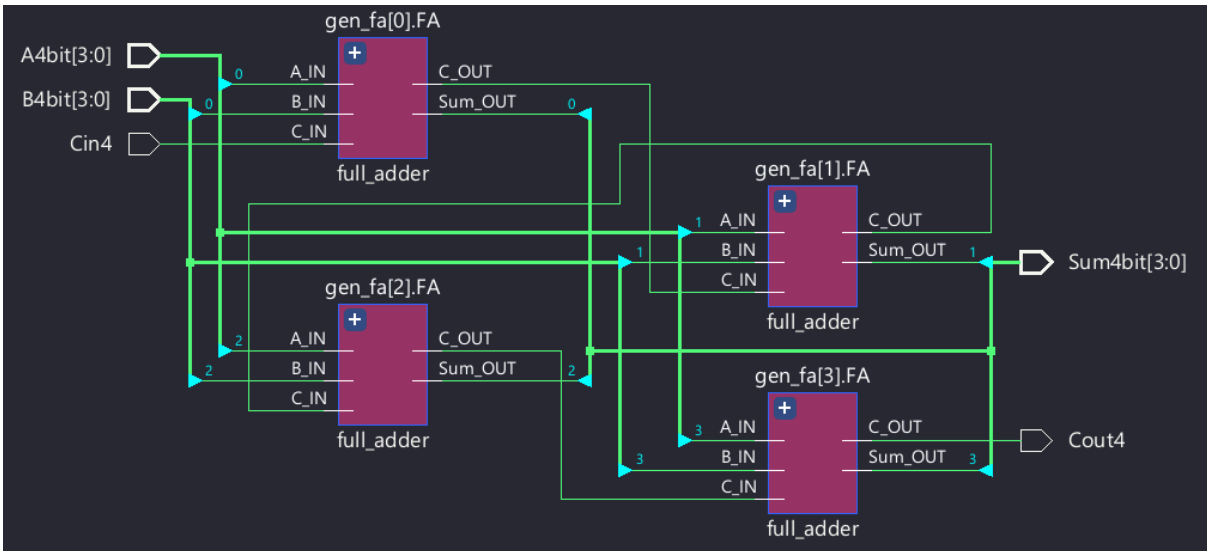
```
Timing Report

Slack:                       inf
  Source:                    B_IN
                               (input port)
  Destination:               C_OUT
                               (output port)
  Path Group:                (none)
  Path Type:                 Max at Slow Process Corner
  Data Path Delay:           7.012ns  (logic 3.968ns (56.592%)  route 3.044ns (43.408%))
  Logic Levels:              3  (IBUF=1 LUT3=1 OBUF=1)

    Location             Delay type                Incr(ns)  Path(ns)    Netlist Resource(s)
  -------------------------------------------------------------------    -------------------
    P10                                               0.000     0.000 r  B_IN (IN)
                         net (fo=0)                   0.000     0.000    B_IN
    P10                  IBUF (Prop_ibuf_I_O)         0.966     0.966 r  B_IN_IBUF_inst/O
                         net (fo=2, routed)           1.370     2.336    B_IN_IBUF
    SLICE_X43Y3          LUT3 (Prop_lut3_I1_O)        0.152     2.488 r  C_OUT_OBUF_inst_i_1/O
                         net (fo=1, routed)           1.674     4.162    C_OUT_OBUF
    P8                   OBUF (Prop_obuf_I_O)         2.850     7.012 r  C_OUT_OBUF_inst/O
                         net (fo=0)                   0.000     7.012    C_OUT
    P8                                                                r  C_OUT (OUT)
  -------------------------------------------------------------------    -------------------
```

Parallel Adder:

- ***VHDL***

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity ParallelAdder is
   port (
      A4bit, B4bit: in std_logic_vector(3 downto 0);
      Cin4: in std_logic;
      Cout4: out std_logic;
      Sum4bit: out std_logic_vector(3 downto 0));
end ParallelAdder;

architecture Structural of ParallelAdder is

   component full_adder is
   port ( A_IN, B_IN, C_IN: in std_logic;
        C_OUT, Sum_OUT: out std_logic);
   end component;

   signal sig_carry: std_logic_vector(4 downto 0);

begin
   -- the compiler turns the first full adder into a half adder (cin 0)
   -- result is efficient hardware with efficient code (dont need to add component ha)

   sig_carry(0) <= Cin4;

   gen_fa: for i in 0 to 3 generate
      FA: full_adder port map (A_IN => A4bit(i),
                         B_IN => B4bit(i),
                         C_IN => sig_carry(i),
                         C_OUT => sig_carry(i+1),
                         Sum_OUT => Sum4bit(i));
   end generate;

   Cout4 <= sig_carry(4);

end Structural;
```

❖ ***RTL Schematic***

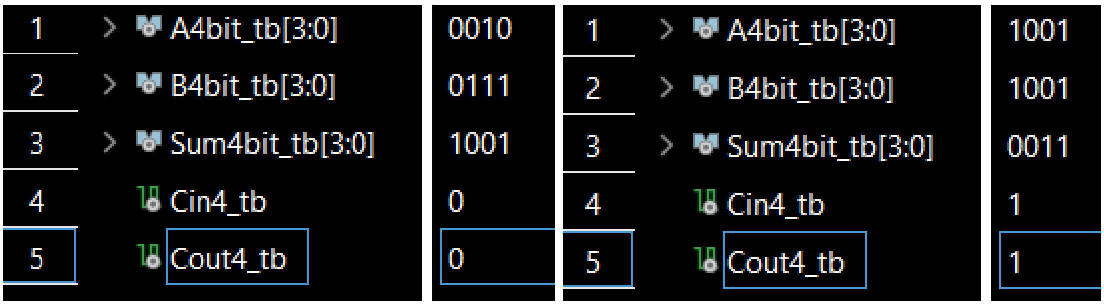


❖ ***Testbench***

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

entity parallel_adder_tb is
end parallel_adder_tb;

architecture Behavioral of parallel_adder_tb is
    component parallel_adder is
        port (
            A4bit, B4bit: in std_logic_vector(3 downto 0);
            Cin4: in std_logic;
            Cout4: out std_logic;
            Sum4bit: out std_logic_vector(3 downto 0));
    end component parallel_adder;

    signal A4bit_tb, B4bit_tb, Sum4bit_tb: std_logic_vector (3 downto 0);
    signal Cin4_tb, Cout4_tb: std_logic;

begin

    DUT: parallel_adder
        port map (A4bit => A4bit_tb,
                  B4bit => B4bit_tb,
                  Cin4 => Cin4_tb,
                  Cout4 => Cout4_tb,
                  Sum4bit => Sum4bit_tb);
    process
```

```
  begin
    for i in 0 to 15 loop
      for j in i to 15 loop -- unique pairs of A, B
        A4bit_tb <= std_logic_vector(to_unsigned(i, 4));
        B4bit_tb <= std_logic_vector(to_unsigned(j, 4));

        Cin4_tb <= '0';
        wait for 20ns;

        assert (unsigned(Sum4bit_tb) = (i+j) mod 16) --ignore Cout
          report "Wrong sum" severity error;

        Cin4_tb <= '1';
        wait for 20ns;

        assert (unsigned(Sum4bit_tb) = (i+j) mod 16) --ignore Cout
          report "Wrong sum" severity error;
      end loop;
    end loop;

    wait;
  end process;
end Behavioral;
```

Η προσομοίωση επιβεβαιώνει την ορθή λειτουργία του παράλληλου αθροιστή. Ενδεικτικά:



❖ *Timing Report*

```
Timing Report

Slack:                     inf
  Source:                  A4bit[0]
                             (input port)
  Destination:             Sum4bit[2]
                             (output port)
  Path Group:              (none)
  Path Type:               Max at Slow Process Corner
  Data Path Delay:         7.949ns  (logic 3.849ns (48.423%)  route 4.100ns (51.577%))
  Logic Levels:            4  (IBUF=1 LUT3=1 LUT5=1 OBUF=1)

    Location             Delay type                Incr(ns)  Path(ns)    Netlist Resource(s)
  -------------------------------------------------------------------    -------------------
    R10                                               0.000     0.000 r  A4bit[0] (IN)
                         net (fo=0)                   0.000     0.000    A4bit[0]
    R10                  IBUF (Prop_ibuf_I_O)         0.969     0.969 r  A4bit_IBUF[0]_inst/O
                         net (fo=3, routed)           1.566     2.535    A4bit_IBUF[0]
    SLICE_X43Y7          LUT5 (Prop_lut5_I2_O)        0.124     2.659 r  Cout4_OBUF_inst_i_2/O
                         net (fo=3, routed)           0.681     3.340    sig_carry_2
    SLICE_X43Y7          LUT3 (Prop_lut3_I0_O)        0.124     3.464 r  Sum4bit_OBUF[2]_inst_i_1/O
                         net (fo=1, routed)           1.853     5.317    Sum4bit_OBUF[2]
    M9                   OBUF (Prop_obuf_I_O)         2.632     7.949 r  Sum4bit_OBUF[2]_inst/O
                         net (fo=0)                   0.000     7.949    Sum4bit[2]
    M9                                                                r  Sum4bit[2] (OUT)
  -------------------------------------------------------------------    -------------------
```

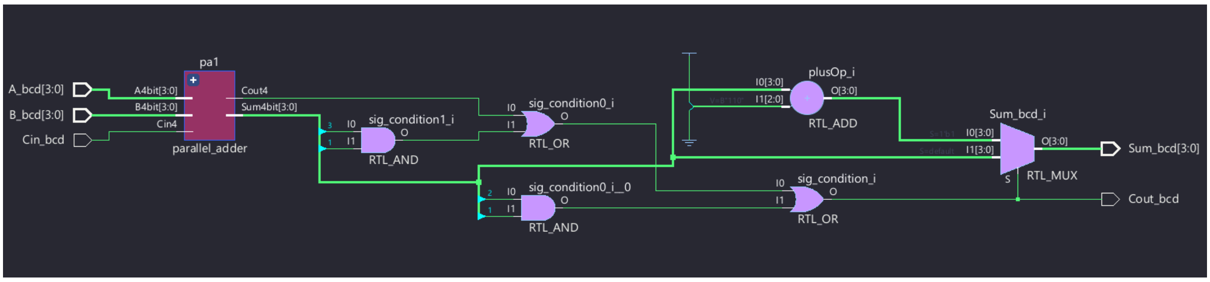
BCD Full Adder:

❖ *VHDL*

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

entity BCD_FA is
    port (A_bcd, B_bcd: in std_logic_vector(3 downto 0);
          Cin_bcd: in std_logic;
          Cout_bcd: out std_logic;
          Sum_bcd: out std_logic_vector(3 downto 0));
end BCD_FA;

architecture Structural of BCD_FA is
    component parallel_adder is
    port (
        A4bit, B4bit: in std_logic_vector(3 downto 0);
        Cin4: in std_logic;
        Cout4: out std_logic;
        Sum4bit: out std_logic_vector(3 downto 0));
    end component;
```

```
    signal sig_tempSum: std_logic_vector(3 downto 0);
    signal sig_carry, sig_condition: std_logic;
begin

    pa1: parallel_adder port map (A4bit => A_bcd,
                                  B4bit => B_bcd,
                                  Cin4 => Cin_bcd,
                                  Cout4 => sig_carry,
                                  Sum4bit => sig_tempSum);

    sig_condition <= sig_carry or (sig_tempSum(3) and sig_tempSum(2)) or (sig_tempSum(3) and
sig_tempSum(1));
    Cout_bcd <= sig_condition;

    Sum_bcd <= std_logic_vector(unsigned(sig_tempSum) + 6) when sig_condition = '1' else
sig_tempSum;
end Structural;
```

❖ ***RTL Schematic***



❖ ***Testbench***

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

entity BCDFA_TB is
end BCDFA_TB;

architecture Behavioral of BCDFA_TB is
    component BCD_FA is
        port (A_bcd, B_bcd: in std_logic_vector(3 downto 0);
              Cin_bcd: in std_logic;
              Cout_bcd: out std_logic;
              Sum_bcd: out std_logic_vector(3 downto 0));
    end component;

    signal A_bcd_tb, B_bcd_tb, Sum_bcd_tb: std_logic_vector(3 downto 0);
```

```
    signal Cout_bcd_tb, Cin_bcd_tb: std_logic;
begin
    DUT: BCD_FA port map (A_bcd => A_bcd_tb,
                          B_bcd => B_bcd_tb,
                          Cin_bcd => Cin_bcd_tb,
                          Cout_bcd => Cout_bcd_tb,
                          Sum_bcd => Sum_bcd_tb);

    process
    begin
        for i in 0 to 9 loop
            for j in i to 9 loop -- unique pairs of A, B
                A_bcd_tb <= std_logic_vector(to_unsigned(i, 4));
                B_bcd_tb <= std_logic_vector(to_unsigned(j, 4));

                Cin_bcd_tb <= '0';
                wait for 20ns;
                Cin_bcd_tb <= '1';
                wait for 20ns;

            end loop;
        end loop;

        wait;
    end process;

end Behavioral;
```

Ενδεικτικά αποτελέσματα της προσομοίωσης:

A_bcd_tb[3:0]	0000	A_bcd_tb[3:0]	0010
B_bcd_tb[3:0]	1010	B_bcd_tb[3:0]	1101
Sum_bcd_tb[3:0]	0001	Sum_bcd_tb[3:0]	0110
Cout_bcd_tb	1	Cout_bcd_tb	1
Cin_bcd_tb	1	Cin_bcd_tb	1

- ***Timing Report***

```
Timing Report

Slack:                    inf
  Source:                 Cin_bcd
                            (input port)
  Destination:            Sum_bcd[2]
                            (output port)
  Path Group:             (none)
  Path Type:              Max at Slow Process Corner
  Data Path Delay:        8.285ns  (logic 4.134ns (49.902%)  route 4.150ns (50.098%))
  Logic Levels:           4  (IBUF=1 LUT5=1 LUT6=1 OBUF=1)

    Location             Delay type                Incr(ns)  Path(ns)    Netlist Resource(s)
  -------------------------------------------------------------------    -------------------
    R8                                                0.000     0.000 r  Cin_bcd (IN)
                         net (fo=0)                   0.000     0.000    Cin_bcd
    R8                   IBUF (Prop_ibuf_I_O)         1.024     1.024 r  Cin_bcd_IBUF_inst/O
                         net (fo=3, routed)           1.605     2.630    Cin_bcd_IBUF
    SLICE_X43Y5          LUT5 (Prop_lut5_I0_O)        0.152     2.782 r  Cout_bcd_OBUF_inst_i_2/O
                         net (fo=4, routed)           0.682     3.463    pa1/sig_carry_2
    SLICE_X42Y5          LUT6 (Prop_lut6_I5_O)        0.326     3.789 r  Sum_bcd_OBUF[2]_inst_i_1/O
                         net (fo=1, routed)           1.863     5.653    Sum_bcd_OBUF[2]
    M9                   OBUF (Prop_obuf_I_O)         2.632     8.285 r  Sum_bcd_OBUF[2]_inst/O
                         net (fo=0)                   0.000     8.285    Sum_bcd[2]
    M9                                                                r  Sum_bcd[2] (OUT)
  -------------------------------------------------------------------    -------------------
```

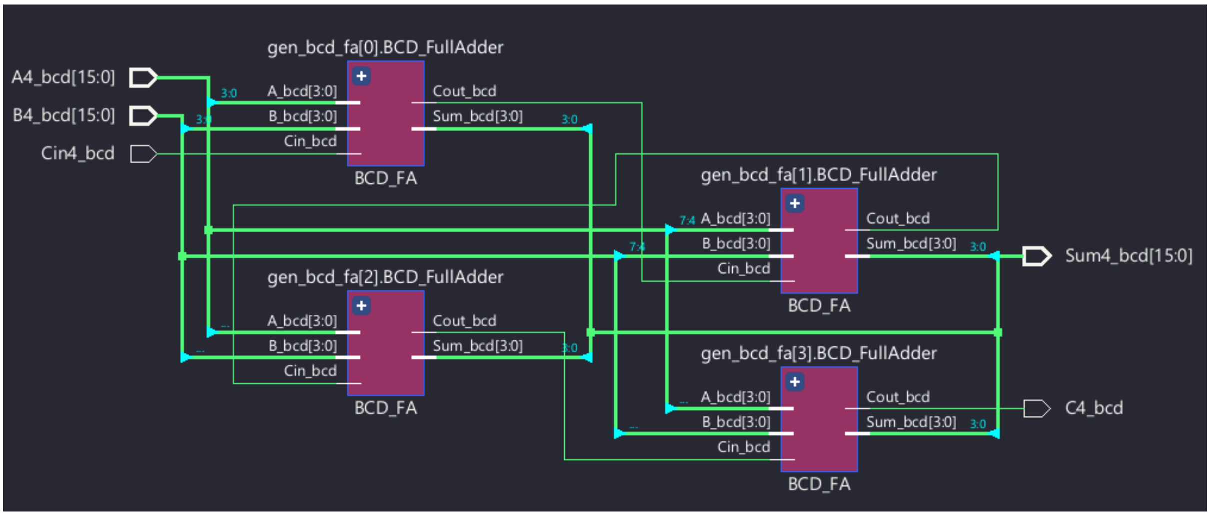
4 BCD Parallel Adder:

- ***VHDL***

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity BCD_PA is
   port (
      A4_bcd, B4_bcd: in std_logic_vector (15 downto 0);
      Cin4_bcd: in std_logic;
      Sum4_bcd: out std_logic_vector (15 downto 0);
      C4_bcd: out std_logic);
end BCD_PA;

architecture Structural of BCD_PA is
   component BCD_FA is
      port (A_bcd, B_bcd: in std_logic_vector(3 downto 0);
          Cin_bcd: in std_logic;
          Cout_bcd: out std_logic;
          Sum_bcd: out std_logic_vector(3 downto 0));
   end component;
```

```
    signal sig_carry: std_logic_vector(4 downto 0);
begin
    sig_carry(0) <= Cin4_bcd;

    gen_bcd_fa: for i in 0 to 3 generate
        BCD_FullAdder: BCD_FA port map
                    (A_bcd => A4_bcd((i*4+3) downto i*4),
                     B_bcd => B4_bcd((i*4+3) downto i*4),
                     Cin_bcd => sig_carry(i),
                     Cout_bcd => sig_carry(i+1),
                     Sum_bcd => Sum4_bcd((i*4+3) downto i*4));
    end generate;

    C4_bcd <= sig_carry(4);
end Structural;
```

❖ *RTL Schematic*



❖ *Testbench*

Έγινε δοκιμή για κάποιες απλές περιπτώσεις και corner cases, καθώς απαιτείται μεγάλη υπολογιστική δύναμη για τον υπολογισμό όλων των δυνατών ζευγών τετραψήφιων δεκαδικών.

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

entity BCD_PA_TB is
end BCD_PA_TB;

architecture Behavioral of BCD_PA_TB is
    component BCD_PA is
```

```
        port (
        A4_bcd, B4_bcd: in std_logic_vector (15 downto 0);
        Cin4_bcd: in std_logic;
        Sum4_bcd: out std_logic_vector (15 downto 0);
        C4_bcd: out std_logic);
    end component;

    signal A4_bcd_tb, B4_bcd_tb, Sum4_bcd_tb: std_logic_vector (15 downto 0);
    signal Cin4_bcd_tb, C4_bcd_tb: std_logic;

begin
    DUT: BCD_PA port map (A4_bcd => A4_bcd_tb,
                    B4_bcd => B4_bcd_tb,
                    Cin4_bcd => Cin4_bcd_tb,
                    Sum4_bcd => Sum4_bcd_tb,
                    C4_bcd => C4_bcd_tb);

    process
    begin
        A4_bcd_tb <= x"1234";
        B4_bcd_tb <= x"5678";
        Cin4_bcd_tb <= '0';

        wait for 20ns;

        A4_bcd_tb <= x"9999";
        B4_bcd_tb <= x"9999";
        Cin4_bcd_tb <= '1';

        wait for 20ns;

        A4_bcd_tb <= x"9999";
        B4_bcd_tb <= x"0001";
        Cin4_bcd_tb <= '1';

        wait for 20ns;

        A4_bcd_tb <= x"0009";
        B4_bcd_tb <= x"0001";
        Cin4_bcd_tb <= '0';

        wait for 20ns;

    end process;
end Behavioral;
```

1	A4_bcd_tb[15:0]	1234
2	B4_bcd_tb[15:0]	5678
3	Sum4_bcd_tb[15:0]	6912
4	Cin4_bcd_tb	0
5	C4_bcd_tb	0

1	A4_bcd_tb[15:0]	9999
2	B4_bcd_tb[15:0]	9999
3	Sum4_bcd_tb[15:0]	9999
4	Cin4_bcd_tb	1
5	C4_bcd_tb	1

1	A4_bcd_tb[15:0]	0009
2	B4_bcd_tb[15:0]	0001
3	Sum4_bcd_tb[15:0]	0010
4	Cin4_bcd_tb	0
5	C4_bcd_tb	0

1	A4_bcd_tb[15:0]	9999
2	B4_bcd_tb[15:0]	0001
3	Sum4_bcd_tb[15:0]	0001
4	Cin4_bcd_tb	1
5	C4_bcd_tb	1

- ***Timing Report***

```
Timing Report

Slack:                    inf
  Source:                 Cin4_bcd
                            (input port)
  Destination:            Sum4_bcd[15]
                            (output port)
  Path Group:             (none)
  Path Type:              Max at Slow Process Corner
  Data Path Delay:        17.580ns  (logic 5.474ns (31.140%)  route 12.106ns (68.860%))
  Logic Levels:           10  (IBUF=1 LUT5=4 LUT6=4 OBUF=1)

    Location             Delay type                Incr(ns)  Path(ns)    Netlist Resource(s)
  -------------------------------------------------------------------    -------------------
    F15                                               0.000     0.000 r  Cin4_bcd (IN)
                         net (fo=0)                   0.000     0.000    Cin4_bcd
    F15                  IBUF (Prop_ibuf_I_O)         0.961     0.961 r  Cin4_bcd_IBUF_inst/O
                         net (fo=4, routed)           2.555     3.517    Cin4_bcd_IBUF
    SLICE_X43Y18         LUT5 (Prop_lut5_I0_O)        0.118     3.635 r  Sum4_bcd_OBUF[3]_inst_i_3/O
                         net (fo=3, routed)           0.819     4.453    Sum4_bcd_OBUF[3]_inst_i_3_n_0
    SLICE_X43Y19         LUT6 (Prop_lut6_I0_O)        0.326     4.779 r  Sum4_bcd_OBUF[5]_inst_i_2/O
                         net (fo=5, routed)           0.746     5.526    Sum4_bcd_OBUF[5]_inst_i_2_n_0
    SLICE_X43Y22         LUT5 (Prop_lut5_I0_O)        0.150     5.676 r  Sum4_bcd_OBUF[7]_inst_i_3/O
                         net (fo=3, routed)           0.668     6.343    Sum4_bcd_OBUF[7]_inst_i_3_n_0
    SLICE_X42Y22         LUT6 (Prop_lut6_I0_O)        0.326     6.669 r  Sum4_bcd_OBUF[8]_inst_i_2/O
                         net (fo=4, routed)           1.042     7.712    Sum4_bcd_OBUF[8]_inst_i_2_n_0
    SLICE_X43Y27         LUT5 (Prop_lut5_I4_O)        0.152     7.864 f  Sum4_bcd_OBUF[11]_inst_i_2/O
                         net (fo=3, routed)           0.812     8.676    Sum4_bcd_OBUF[11]_inst_i_2_n_0
    SLICE_X43Y29         LUT6 (Prop_lut6_I2_O)        0.326     9.002 r  Sum4_bcd_OBUF[12]_inst_i_2/O
                         net (fo=4, routed)           0.947     9.949    Sum4_bcd_OBUF[12]_inst_i_2_n_0
    SLICE_X43Y31         LUT5 (Prop_lut5_I4_O)        0.152    10.101 r  Sum4_bcd_OBUF[15]_inst_i_3/O
                         net (fo=4, routed)           0.823    10.924    Sum4_bcd_OBUF[15]_inst_i_3_n_0
    SLICE_X43Y33         LUT6 (Prop_lut6_I1_O)        0.326    11.250 r  Sum4_bcd_OBUF[15]_inst_i_1/O
                         net (fo=1, routed)           3.693    14.943    Sum4_bcd_OBUF[15]
    F14                  OBUF (Prop_obuf_I_O)         2.637    17.580 r  Sum4_bcd_OBUF[15]_inst/O
                         net (fo=0)                   0.000    17.580    Sum4_bcd[15]
    F14                                                               r  Sum4_bcd[15] (OUT)
  -------------------------------------------------------------------    -------------------
```